Experimental Comparison of 2-bit Vector Quantizers

Diego Manuel Maldonado Nunez
Student ID: 202620726

July 2026

1 Introduction

Vector quantization is a compression method in which a real-valued vector is represented by an index into a finite codebook. Instead of storing every coordinate of the original vector using full precision, the quantizer maps the vector to one of a limited number of representative vectors, and the dequantizer maps the stored code back to an approximate reconstruction. This is useful when the original vectors are expensive to store or move, but some approximation error is acceptable.

This experiment studies vector quantization in a controlled 24-dimensional setting. Each algorithm receives the same input vectors and is constrained to an effective rate of 2 bits per dimension. The goal is not only to determine which method reconstructs vectors most accurately, but also to determine the practical cost of using each method. For this reason, the experiment evaluates three dependent variables: reconstruction quality, quantization time, and dequantization time.

The four algorithms compared are:

- **Leech Lattice Vector Quantization**: a lattice-based vector quantizer built on the 24-dimensional Leech lattice. The experiment uses 24-dimensional input blocks because the Leech lattice is defined in $\mathbb{R}^{24}$, so each block can be treated as one lattice-quantized vector. This dimension is also mathematically important: the Leech lattice achieves the optimal sphere packing density in 24 dimensions [5], and recent Leech Lattice Vector Quantization work uses its symmetry, shell structure, indexing, and search procedures to quantize without materializing a full unstructured codebook [4]. In this experiment, the finite codebook is the union of Leech lattice shells up to shell 13. Shell 13 is used because the cumulative shell count requires 48 index bits, which gives exactly $48/24 = 2$ bits per dimension.

- **Uniform scalar 2-bit**: a simple coordinate-wise quantizer. Each coordinate is rounded independently to one of a small number of uniformly spaced levels. This method is simple and fast, but it does not use relationships between coordinates.

- **QTIP bitshift**: an additional quantization algorithm based on the bitshift codebook component from QTIP, which stands for Quantization with Trellises and Incoherence Processing [1]. QTIP uses trellis-coded quantization to support very high-dimensional quantization, and introduces a hardware-efficient bitshift trellis for fast decoding.

- **QuIP# E8P12**: an additional 2-bit vector quantizer based on the QuIP# E8 lattice codebook [2]. The E8P12 codebook quantizes 8-dimensional blocks with 16-bit indices, so each 24-dimensional experimental block is represented by three E8P12 indices for an effective rate of 2 bits per dimension.

The main research question is:

> Which 2-bit quantization algorithm gives the best trade-off between reconstruction quality, quantization time, and dequantization time?

The expected trade-off is that a structured vector quantizer such as Leech Lattice Vector Quantization may achieve better reconstruction quality, while simpler scalar or bitshift methods may run faster.

2 Experiment Design

2.1 Variables

The independent variable is the quantization algorithm, with four levels: Leech Lattice Vector Quantization, Uniform scalar 2-bit, QTIP bitshift, and QuIP# E8P12. The dependent variables are SQNR bits, quantize seconds, and dequantize seconds.

2.1.1 SQNR bits

SQNR bits is the signal-to-quantization-noise ratio expressed in bits. Higher is better. This is the quality metric used for sample-size calculation, hypothesis testing, and interpretation. For an original vector $x \in \mathbb{R}^d$ and its dequantized reconstruction $\hat{x} \in \mathbb{R}^d$, SQNR bits measures the ratio between signal power and quantization-noise power, expressed in base-2 logarithmic units:

$$\mathrm{SQNR}_{\mathrm{bits}}(x, \hat{x}) = \frac{1}{2}\log_2\left(\frac{\frac{1}{d}\sum_{i=1}^{d} x_i^2}{\frac{1}{d}\sum_{i=1}^{d}(x_i - \hat{x}_i)^2}\right).$$

This formulation is useful because it normalizes the reconstruction error by the power of the original vector.

2.1.2 Quantize seconds

Quantize seconds is the wall-clock time needed to encode one vector. Lower is better. This measures the cost of producing the compressed representation, which matters when vectors must be encoded online or repeatedly during a larger pipeline.

2.1.3 Dequantize seconds

Dequantize seconds is the wall-clock time needed to decode one vector. Lower is better. This measures the cost of reconstructing vectors from their compressed representation, which matters when compressed vectors are read frequently or used in downstream computation.

These three metrics were selected because they correspond to the practical trade-off in a quantization system: reconstruction quality, encoding cost, and decoding cost. Considering them together makes it possible to distinguish an algorithm that is accurate but expensive from one that is faster but less accurate.

2.2 Hypotheses

For each metric, the global ANOVA-style hypothesis is:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$$

$$H_1 : \exists i, j \text{ such that } \mu_i \neq \mu_j$$

For SQNR bits, larger means better reconstruction. For quantization and dequantization time, smaller means better runtime performance.

3 Data Collection Protocol

The experiment was implemented in `quantizer_sample_size_experiment.ipynb`. All input vectors were actual model-weight blocks from Qwen3-0.6B [3]. To avoid bias from using only the first layer, blocks were sampled randomly without replacement from the MLP down-projection tensors across the transformer layers, matching the pattern `model.layers.*.mlp.down_proj.weight`. The random seed was fixed to 0, and the weight-sampling seed was fixed to 2026 for repeatability.

The Leech lattice is intrinsically 24-dimensional, so each experimental block used 24 weights. Since the quantizers in this experiment do not store an explicit gain or scale value, each 24-dimensional weight block was normalized by its RMS before quantization. This keeps the relative shape of the real model-weight block while matching the scale assumptions of the 2-bit codebooks.

The experiment was conducted in two phases:

1. **Pilot experiment**: 30 paired blocks were collected. In each block, the same vector was evaluated by all four algorithms.

2. **Final experiment**: the final number of paired blocks was chosen using the pilot-based sample-size calculation described below. The final sample covered 67 weight blocks from 24 distinct down-projection layers.

Each block consisted of one vector evaluated by every algorithm. This creates a paired design: all algorithms are compared on the same input vectors. The order of algorithms was randomized during timing to reduce ordering effects. Before timing, each algorithm was warmed up for 5 iterations. For GPU timing, CUDA synchronization was performed before and after each measured call, so asynchronous CUDA execution would not contaminate timing measurements.

Each CSV row records the phase, block id, sampled weight tensor, layer, tensor-local block index, method, SQNR bits, quantization time, dequantization time, the original vector, the dequantized vector, dimension, and seed. The final data used in this report is:

```
results/quantizer_sample_size_experiment/seed0_llm_weight_blocks/final_experiment.csv
```

4 Data Analysis Protocol

First, the pilot data was used to estimate the within-algorithm variance for each metric. A power calculation for a four-algorithm mean comparison was then performed. Two effect scenarios were evaluated:

- **Two levels symmetric**: two algorithms differ symmetrically around the grand mean.

- **One versus rest**: one algorithm differs from the other algorithms.

For each metric and scenario, the minimum sample size per algorithm was calculated using the noncentral F distribution. The final experiment used the largest required sample size across all metrics and scenarios. The target significance level was $\alpha = 0.05$ and the desired power was 0.80.

The minimum interesting differences were:

- SQNR bits: 0.10 bits. This is small relative to the absolute SQNR level, but large enough to indicate a meaningful reconstruction-quality shift between algorithms.
- Quantize time: 0.001 seconds. Encoding is the slower runtime operation in this benchmark, so a one-millisecond difference per vector would be practically noticeable when many vectors are quantized.
- Dequantize time: 0.00005 seconds. Decode calls are short, so 50 microseconds is a meaningful latency change.

After collecting the final data, descriptive statistics were computed for each metric and algorithm, including mean, standard deviation, coefficient of variation, and 95% confidence intervals. The global tests used were repeated-measures ANOVA tests. The normality assumption was checked using Shapiro-Wilk tests on repeated-measures ANOVA residuals. Pairwise post-hoc comparisons were performed with Bonferroni correction over all planned metric comparisons.

4.1 Available Hardware and Software

The experiment was executed on the following machine:

- CPU: AMD Ryzen 9 5950X 16-Core Processor, 16 cores / 32 threads.
- RAM: 62 GiB system memory.
- GPU: NVIDIA GeForce RTX 3080, 10 GiB VRAM, compute capability 8.6.
- NVIDIA driver: 595.80.
- Operating system: Linux 7.0.13-200.fc44.x86_64.
- Python: 3.12.9.
- PyTorch: 2.12.1+cu130, CUDA 13.0 available.
- NumPy: 2.4.6, Pandas: 3.0.3, SciPy: 1.18.0.

5 Actual Experiment

5.1 Sample Size Calculation

Table 1 shows the pilot-based sample-size calculation on randomly sampled Qwen3-0.6B weight blocks. The largest required sample size was 67 observations per algorithm, coming from the SQNR metric under the two-level symmetric scenario. Therefore, the final experiment used 67 paired blocks.

This sample size is small relative to the full model because Qwen3-0.6B contains many millions of weights. Evaluating every possible 24-dimensional block would turn the study into a much larger full-checkpoint benchmark and would be expensive for the exact Leech lattice quantizer and the QTIP baseline. The study therefore uses a pilot-powered random sample of blocks as a controlled estimate of behavior on model weights, rather than a full census of the checkpoint.

Table 1: Pilot-based sample-size calculation.

Metric	Scenario	Pilot SD	Min. diff.	Required n
SQNR bits	Two levels symmetric	0.173003	0.10000	67
SQNR bits	One versus rest	0.173003	0.10000	45
Quantize seconds	Two levels symmetric	0.000253	0.00100	3
Quantize seconds	One versus rest	0.000253	0.00100	3
Dequantize seconds	Two levels symmetric	0.000054	0.00005	27
Dequantize seconds	One versus rest	0.000054	0.00005	19

5.2 Descriptive Results

The final experiment collected 67 paired Qwen3-0.6B weight blocks for each algorithm. These blocks were randomly sampled from 24 distinct MLP down-projection layers. Figure 1 shows the distributions of the main metrics.

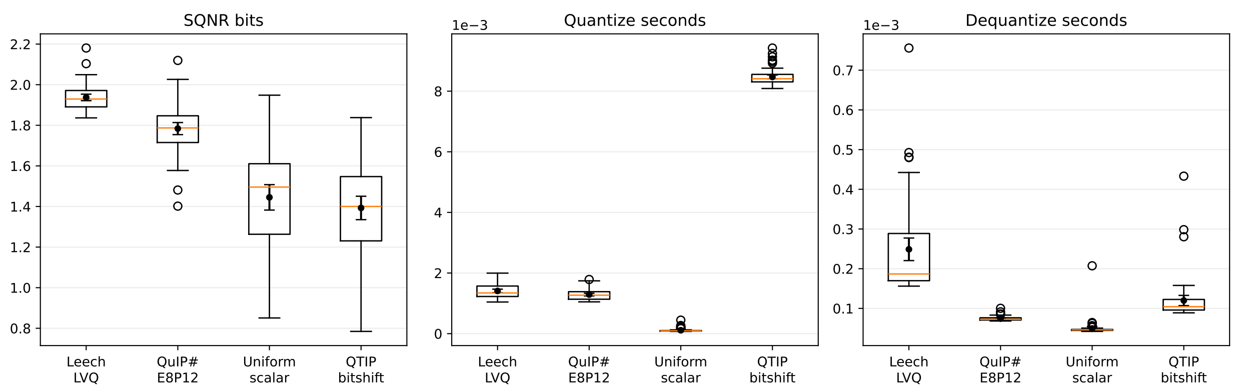

Figure 1: Final metric distributions for the four algorithms. Black points show the mean with 95% confidence intervals.

Table 2 summarizes reconstruction quality. Leech Lattice Vector Quantization achieved the best reconstruction quality, with the highest SQNR bits, followed by QuIP# E8P12.

Table 2: Final reconstruction quality statistics.

Algorithm	SQNR mean	SQNR SD	SQNR 95% CI	Coef. variation
Leech Lattice Vector Quantization	1.937313	0.065726	[1.921281, 1.953345]	0.033927
QuIP# E8P12	1.783790	0.121918	[1.754052, 1.813528]	0.068348
Uniform scalar 2-bit	1.444994	0.256908	[1.382329, 1.507659]	0.177791
QTIP bitshift	1.392786	0.236477	[1.335105, 1.450468]	0.169787

Table 3 summarizes runtime. Uniform scalar quantization was the fastest method for both quantization and dequantization.

Table 3: Final runtime statistics.

Algorithm	Quantize mean (s)	Quantize SD	Quantize 95% CI	Dequantize mean (s)	Dequantize SD	Dequantize 95% CI
Uniform scalar 2-bit	0.000112	0.000059	[0.000098, 0.000127]	0.000050	0.000020	[0.000045, 0.000054]
QuIP# E8P12	0.001289	0.000183	[0.001244, 0.001333]	0.000074	0.000005	[0.000073, 0.000076]
Leech Lattice Vector Quantization	0.001409	0.000241	[0.001350, 0.001468]	0.000249	0.000116	[0.000220, 0.000277]
QTIP bitshift	0.008469	0.000260	[0.008406, 0.008533]	0.000120	0.000053	[0.000107, 0.000133]

5.3 Statistical Tests

The repeated-measures ANOVA tests rejected the null hypothesis for all three primary metrics. Table 4 shows strong evidence of algorithm differences.

Table 4: Global statistical tests on final data.

Metric	Test	Statistic	p-value
SQNR bits	Repeated-measures ANOVA	164.90	1.37×10^{-53}
Quantize seconds	Repeated-measures ANOVA	27469.27	4.34×10^{-259}
Dequantize seconds	Repeated-measures ANOVA	124.96	1.91×10^{-45}

The post-hoc comparisons showed the following ordering:

- **SQNR bits**: Leech Lattice Vector Quantization > QuIP# > Uniform and QuIP# > QTIP. Uniform had a higher mean than QTIP, but that pairwise difference was not significant after Bonferroni correction.
- **Quantize time**: Uniform < QuIP# < Leech Lattice Vector Quantization < QTIP.
- **Dequantize time**: Uniform < QuIP# < QTIP < Leech Lattice Vector Quantization.

Seventeen of the eighteen pairwise comparisons were significant after Bonferroni correction. The adjusted alpha was 0.002778. The only non-significant comparison was Uniform scalar 2-bit versus QTIP bitshift for SQNR bits.

5.4 Normality Check

Table 5 summarizes the Shapiro-Wilk checks on repeated-measures ANOVA residuals. SQNR residuals passed the normality check, but quantization-time and dequantization-time residuals did not. Therefore, the repeated-measures ANOVA results for timing should be interpreted with caution.

Table 5: Shapiro-Wilk residual normality checks.

Metric	Test	p-value	Passes $\alpha = 0.05$
SQNR bits	Shapiro-Wilk residual normality	0.8389	Yes
Quantize seconds	Shapiro-Wilk residual normality	0.000107	No
Dequantize seconds	Shapiro-Wilk residual normality	6.66×10^{-16}	No

6 Discussion

The results show that there is no single best algorithm across all metrics for Qwen3-0.6B weight blocks. Leech Lattice Vector Quantization is the best choice when reconstruction quality is the priority. It achieved the highest SQNR and was significantly better than QuIP#, scalar quantization, and QTIP bitshift. QuIP# E8P12 was the second-best method for reconstruction quality, significantly improving on both scalar quantization and QTIP bitshift. However, the higher-quality vector quantizers came with higher runtime cost. Uniform scalar quantization was fastest for both quantization and dequantization, while QuIP# was second fastest for both timing metrics. QTIP bitshift did not improve SQNR in this configuration: its SQNR mean was slightly below uniform scalar quantization, and the SQNR difference between those two methods was not significant after correction.

The random multi-layer sampling reduces the risk of first-layer bias, but the sample remains intentionally small compared with the full checkpoint. The results should therefore be interpreted as a controlled estimate on representative weight blocks, not as a complete full-model quantization benchmark.

The Shapiro-Wilk check is an important limitation. Quantization-time and dequantization-time residuals violate the normality assumption, so the repeated-measures ANOVA results for timing should be interpreted cautiously. However, the timing effects are large and the descriptive ordering is clear. The exact ANOVA p-values for timing should not be over-interpreted.

7 Conclusion

This experiment compared four 2-bit quantization algorithms on actual Qwen3-0.6B weight blocks under a controlled paired design. The final sample size was calculated from a random multi-layer weight-block pilot study, resulting in 67 paired observations.

Leech Lattice Vector Quantization produced the best reconstruction quality, with SQNR mean 1.937 bits. QuIP# E8P12 was second best on reconstruction quality, with SQNR mean 1.784 bits. Uniform scalar quantization was the fastest method, with mean quantization time 0.000112 seconds and mean dequantization time 0.000050 seconds. QTIP bitshift did not dominate either quality or runtime in this experimental configuration.

Therefore, the recommended algorithm depends on the objective. If quality is the priority, Leech Lattice Vector Quantization should be selected. If runtime is the priority, especially for quantization and dequantization latency, the uniform scalar quantizer is preferable. Under these measurements, no algorithm is best simultaneously for SQNR, quantization time, and dequantization time.

References

[1] Albert Tseng, Qingyao Sun, David Hou, and Christopher De Sa. *QTIP: Quantization with Trellises and Incoherence Processing*. arXiv:2406.11235, 2024. `https://arxiv.org/abs/2406.11235`

[2] Albert Tseng, Jerry Chee, Qingyao Sun, Volodymyr Kuleshov, and Christopher De Sa. *QuIP#: Even Better LLM Quantization with Hadamard Incoherence and Lattice Codebooks*. Proceedings of the 41st International Conference on Machine Learning, 2024. `https://arxiv.org/abs/2402.04396`

[3] Qwen Team. *Qwen3-0.6B*. Hugging Face model repository, 2025. https://huggingface.co/Qwen/Qwen3-0.6B

[4] Tycho F. A. van der Ouderaa, Mart van Baalen, Paul Whatmough, and Markus Nagel. *Leech Lattice Vector Quantization for Efficient LLM Compression.* arXiv:2603.11021, 2026. https://arxiv.org/abs/2603.11021

[5] Henry Cohn, Abhinav Kumar, Stephen D. Miller, Danylo Radchenko, and Maryna Viazovska. *The sphere packing problem in dimension 24*. Annals of Mathematics, 185(3), 2017. https://annals.math.princeton.edu/2017/185-3/p08